Prerequisites:

- To install Batchalign, follow the instructions at https://talkbank.github.io/batchalign/ You can use either the GUI Application (Mac, Windows, or Linux) or the CLI (command line interface) version.

1. Next, you will need to prepare your audio or video file and put it inside any folder. Batchalign works on most major audio format types, but we'd recommend .wav, .mp3, and .mp4.
2. You can put as many files as you wish into your input folder, and they will be processed in sequence. If your machine has enough memory and multiple processor cores, as with the M2 Apple MacStudio, it will detect the additional CPUs and process multiple files in parallel. Nested folders (folder that contains folders of audio) are supported in the expected way.
3. Batchalign supports different processes with different verbs. The three most used are marked with an asterisk:
    - ***transcribe** provides transcription directly from audio or video. This only requires raw media files (audio or video) in your input folder.
    - ***align** produces utterance- and word-level alignment of a text when you place both the media and transcript files into /input. If utterance bullets are present, Batchalign will use them even if they are wrong, potentially worsening the alignment of the whole file; hence, it is best to first remove current bullets using this CLAN command: *chstring -cbullets.cut *.cha* +1 unless you are sure the bullets are absolutely correct. It uses the @Languages line in the transcript to detect what language model to use.
    - ***morphotag** uses Stanza, following Universal Dependencies, to add %mor and %gra lines to a transcript. This function does not require a media file. It uses the @Languages line in the transcript to detect what language model to use.
    - **version** lists the version of batchalign.
4. Each of the command verbs has some additional switches that modify usage. To see the complete list of switches for a given command, use the --help flag following the verb. For instance, for the transcribe verb, you would type the command in this format: batchalign transcribe --help. Understanding these additional switches is particularly important for the transcribe verb.
5. You can use either Whisper or Rev-AI for transcription. The default mode for English uses Rev-AI. For this, you will need to open a rev.ai account. Rev-AI provides you with 6 free hours for your new account. Charges are $.02/minute of audio for this service. Go to rev.ai, sign up, and on the left side of your dashboard, you will find a tab called Access Token. Click generate to generate a new token, copy and paste the key to somewhere you can find later. If you want to comply with IRB rules against sending data to third parties, you can configure your Rev-AI service to auto-delete your data after processing, as described at this page. It is also possible to use Rev-AI for data that must be HIPAA compliant, as described at this page.
6. In addition to Rev.AI, we have adjusted Whisper, a local ASR model, to perform nearly as well as Rev-AI. Whisper has a wider potential coverage of languages than Rev-AI. Also, Whisper seems to be better than Rev-AI for Spanish. However, for all of the other languages

that it supports, Rev-AI does better than Whisper. Moreover, Rev-AI can create reasonably useful speaker diarization and Whisper does not do this at all. Although Whisper runs much more slowly than Rev-AI and requires a minimum memory of about 16GB, some projects may prefer Whisper's local mode of operation.

7. The first time you run Batchalign, the program will take about 5 minutes to download the material that will go into various cache folders on your system. After that, the system will ask you for your Rev.AI key from step 6 above, which you will need to paste into the program when asked if you wish to use Rev.AI. Cut and paste that from the place where you saved it earlier.
8. Batchalign also supports four systems for Chinese languages: FunASR for child Cantonese (fast), Qwen3 for child Cantonese (slow), Paraformer-zh for child Mandarin, and Tencent for adult Chinese languages. The first three are free, but Tencent requires a license and API code.
9. If you are using the GUI Application, you can use the radio buttons to select a language and an ASR system. You can select the folder to be processed by dragging it into the processing window.
10. If you are using the CLI version, the command for transcribing is:
    - batchalign transcribe --lang=[3 letter ISO language code] [folder]
    - For Rev.AI: batchalign transcribe --lang=eng [folder]
    - For Whisper (offline): batchalign transcribe --lang=eng --engine=openai [folder]
11. The program will provide output as it processes each input file, and you will soon see transcribed or coded CHAT (*.cha) files in your output folder(s)!

*Only the **transcribe** function requires the --lang flag. All other functions will read language information from the input CHAT file.*

Support Information:
Please feel free to reach out if you have questions! You can send email to macw@cmu.edu or to houjun@cmu.edu. Should you reach out for help, please run “batchalign version” to tell us which version you are using.